

ADITHYA J

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PROFESSIONAL SUMMARY

B.Tech student pursuing Computer Science Engineering (AI & ML) at VIT Bhopal, with hands-on experience in end-to-end machine learning pipelines, computer vision, NLP, and cloud deployment. Proficient in model training, data preprocessing, model evaluation, and deployment using Python, TensorFlow, PyTorch, and Microsoft Azure. Targeting SDE and Cloud Engineering roles.

TECHNICAL SKILLS

Languages:	Python, Java, C++, JavaScript
ML & AI:	Machine Learning, Deep Learning, NLP, Computer Vision, ML Pipelines, Model Deployment, Evaluation Metrics, YOLOv8, Reinforcement Learning
Frameworks:	TensorFlow, PyTorch, scikit-learn, Ultralytics, OpenCV, MediaPipe, NumPy, Pillow
Cloud & DevOps:	Microsoft Azure, AWS, Docker, REST APIs
Databases:	MongoDB, MySQL
Tools:	Git, GitHub, Jupyter Notebook, VS Code, OpenAI Gym
Web:	ReactJS, HTML, CSS

PROJECTS

Smart Navigation System for Visually Impaired

Python | YOLOv8 | PyTorch | OpenCV | Pillow | NumPy | pyttsx3 | Threading | Laptop / Edge Device

- Real-time object detection pipeline using YOLOv8 with confidence thresholding and Non-Maximum Suppression (NMS); spatial position logic (left / center / right) and distance approximation (near / far / very near) for scene understanding.
- Offline Text-to-Speech (TTS) voice alerts via pyttsx3 with threading for non-blocking audio output; persistent logic prevents repeated alerts for the same detected object.

Real-Time Posture Tracking System

Python | OpenCV | MediaPipe | NumPy | Git

- Real-time inference pipeline with keypoint detection achieving 90%+ accuracy at 28 FPS; body angle deviation overlays with sub-200ms feedback latency.

Hierarchical Intent Classification for Financial Service Queries

Python | NLP | Naive Bayes | Bag-of-Words (BoW) | Small Language Model (SLM) | Banking77 Dataset

- Lightweight NLP pipeline built from scratch using SLM principles — classical probability and rule-based morphology — classifying 77 banking intents without Transformer models; hierarchical layers resolve semantic overlap between similar intents.

Cart-Pole Balancing using Reinforcement Learning

Python | OpenAI Gym | Reinforcement Learning | NumPy

- RL agent trained via exploration-exploitation strategy in OpenAI Gym; evaluated using reward-per-episode metrics, average reward curves, and learning curve analysis.

EDUCATION

Vellore Institute of Technology, Bhopal

B.Tech in Computer Science and Engineering (AI & ML) — CGPA: 8.39 / 10.0

2023 – 2027

Navy Children School, Kochi

Higher Secondary Certificate (HSC) — 90.6%

2022 – 2023

Kendriya Vidyalaya No.2, Port Blair

Secondary School Certificate (SSC) — 94.2%

2020 – 2021

POSITIONS OF RESPONSIBILITY

General Secretary — Feedbox College Club, VITB

2025

- Led a team of 80+ members; planned and executed 3 major events with strong operational coordination.

Corporate & Outreach — E-Cell, VITB

2024 – 2025

- Used data-driven outreach strategies for brand partnerships and PR collaborations; contributed to 6 successful events.

PR & Outreach — Meraki Club, VITB

2023 – 2025

- Managed PR campaigns securing brand and sponsor partnerships; coordinated execution of 9 major events.